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Managing the transition to sustainability in an emerging economy: Evaluating green growth policies in Indonesia

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ABSTRACT

Rapidly growing emerging economies, such as Brazil, China, India and Indonesia have joined the ranks of major industrialized countries as significant emitters of greenhouse (GHG). Their rapid economic growth has, however, brought about considerable environmental degradation, health costs, material damage, and a decrease in agriculture productivity. The notion of "green growth" has been proposed as a way out. This paper examines the case of Indonesia and evaluates its search for green growth through an approach that combines a so-called Energy Mix Policy and a REDD+ program.

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1. Introduction

One of the greatest challenges of the current century is climate change, caused by Greenhouse Gas (GHG) emissions arising from various human activities since the beginning of the Industrial Revolution. Major emitters of GHGs are rapidly growing economies, particularly emerging economies like Brazil, China, India, and Indonesia. China's annual economic growth of 9–10% requires a great deal of energy, most of which derived from fossil fuels. This has resulted in considerable environmental

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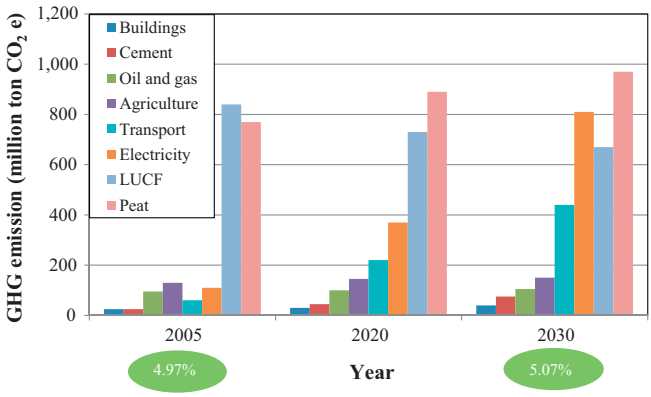

Fig. 1. GHG emissions in 2005 and projected for 2020 and 2030.
Indonesia National Climate Change Commission (2010).

degradation and health costs, a decrease in agricultural productivity and material damage. These costs have been estimated to be equivalent to half of China's annual economic growth (Clark, 2009). To overcome this unfavourable development, "green growth" has been proposed as a new way of pursuing economic growth and development, which purportedly avoids environmental degradation, loss of biodiversity, and unsustainable use of natural resources (OECD, 2011). It is emphasized at the outset that in order for green growth to be inclusive and equitable, growth and development policies should pay equal attention to the sustainable development triad society, economy and environment. Several key strategic elements are identified to achieve green growth: phasing out oil subsidies, acceleration of innovation and diffusion of green technologies, support a sustainability transition through policies aimed at upgrading skills and capabilities of workers, strengthening of international cooperation, and measuring progress by development of a transparent and responsible accounting framework involving social, economic and environment indicators.

The concept of "green growth" has been connected to the "green economy for sustainable development and poverty reduction", a theme of the Rio+20 United Nations Conference on Sustainable Development (United Nations, 2011). The transition to green growth entails a shift from fossil fuels to renewable energy sources. This in turn involves many innovations which must be analysed in an integrated way and which have to incorporate adequate technological, policy and social responses to achieve global sustainability (Reid et al., 2010).

Against the background the aim of green growth, we examine how transition processes aimed at sustainability are currently being addressed in Indonesia, one of the major emerging economies in the world. Indonesia emitted about 2.1 Gt CO_2e in 2005, which is equivalent to 4.97% of global GHG emission as shown in Fig. 1, while it contributed only 0.6% to the global Gross Domestic Product (GDP). Unlike China and India, most of Indonesia's GHG emissions do not originate from industrial activities, but are mainly due to peat fires and deforestation (Indonesia National Climate Change Commission, 2010). Not surprisingly, Indonesia plays an increasingly important role in the negotiations and institutions for an international "climate regime".

2. Energy and climate policy

With a population of 230 million people in 2009, Indonesia is the country with the fourth largest population in the world. It is also home to the third largest tropical forest. Indonesia covers 790×10^6 ha, with a land territory of about 200×10^6 ha, of which two thirds is forest. While its population increased by 1.3% per year between 2001 and 2005, its GDP grew at 5.6% per year in the last 5 years. Energy demand increased rapidly at a rate of 3.7% per year between 1990 and 2008, and over 50% of the energy demand is met by fossil oil. Depleting oil reserves, lack of investment in oil exploration and an imbalance between oil consumption and production changed Indonesia's status in

2004 from net oil exporter to net oil importer (Indonesia Ministry of Energy and Mineral Resources, 2006). In 2005, due to a hike in oil prices, the oil subsidy provided by the government of Indonesia jumped from 2 billion US$ to 9.7 billion US$, equal to 3.4% of the nation's GDP (World Bank, 2005). Fossil fuels made up 96% of Indonesia's energy – 56% oil, 19% coal and 21% gas – while renewable energy made up the remaining 4% – of which 3% was hydropower and 1% geothermal energy.

In 2006, the government of Indonesia implemented the so-called Energy Mix Policy aimed at reducing oil consumption by partially shifting to renewable energy sources. Under this policy, the share of renewable sources in the total energy mix is foreseen to increase from 4% in 2006 to 15% in 2025: 5% from geothermal, 5% from biofuels and another 5% from other renewable like wind and solar (Indonesia Ministry of Energy and Mineral Resources, 2006). Increased use of biofuels, mostly in the transportation sector, is expected to create jobs to alleviate poverty, to generate income as an export commodity besides leading to a net reduction in GHG emissions (Jupesta, 2011). In 2010, to accelerate the shift to clean energy, the National Energy Agency proposed to enhance the share of renewable energy to 25% in 2025, in a plan called "Vision 25/25". However, this proposal is still under discussion between the government and the parliament and not yet formalized into legislation (Indonesia National Energy Agency, 2011).

While at the global level 57% of GHG emissions originate from fossil fuel combustion, at the national scale 62% of such emissions are caused by peat fires and changes in land use and forestry (LUCF) (Boer et al., 2010). The total GHG emissions are estimated to grow from 2.1 to 3.3 $GtCO_2$ between 2005 and 2030 under a business-as-usual (BAU) scenario. GHG emissions from peat fire and LUCF are expected to grow from 0.91 $GtCO_2$ to 1.64 GtCO2 in the same period. The 13th session of the Conference of the Parties (COP) of the United Nations Framework Convention on Climate Change (UNFCCC), held in 2007, agreed to explore the policies and financial incentives that would reduce emissions from deforestation and forest degradation (REDD). Realizing that GHG emissions in Indonesia mostly come from peat fires and LUCF, the government of Indonesia made a path-breaking announcement at the COP 15th UNFCCC in Copenhagen in 2009 that the country will reduce its GHG emissions by 26% in 2020 compared to the BAU scenario, and by a further 15% with adequate international support. In January 2010, the government prepared a National Action Plan on Climate Change. Later, in May 2010, the government of Norway and the government of Indonesia agreed on a REDD+ Partnership Program, in which Norway will grant 1 billion US$ for the REDD+ Partnership Program if certain forest governance, monitoring and financing objectives are to be achieved (Solheim, 2010). Still, implementation of this partnership program is facing delay due to the complexity of the carbon accounting system needed to verify progress in reducing GHG emissions from peat fire and LUCF.

3. Policy impacts

There are three key impacts of these energy and climate policies. First, the Energy Mix Policy had a positive impact on the environment by reducing subsidies on fossil fuels from 3.4% to 1.8% of the nation's GDP in 2009 (OECD, 2010). According to the United Nations Environment Programme (UNEP), eliminating subsidies for fossil fuels worldwide could reduce global GHG emissions by as much as 6% and add 0.1% to global GDP (UNEP, 2010). Recently, the government has started an effort to shift the use of kerosene for household cooking, which is highly subsidized, to liquefied petroleum gas (LPG). The phase out of kerosene is scheduled to come into effect by 2012 and of other subsidized fuels by 2014. This removal of subsidies is in line with the government's energy policy, aimed at conserving energy by using fewer polluting sources (shifting from oil to LPG or to biofuels) and to change consumer behaviour towards more efficient energy use. The government provides a 'Direct Cash Transfer' program to help the poor adapt to the increasing price of energy. Moreover, oil subsidies in the transportation sector will be stopped for private vehicles starting from 2011, since most of these subsidies go anyway to the rich rather than to the poor (OECD, 2010).

A second key impact of the energy and climate policies has been clearing and expansion of land for production of biofuel crops. This process picked up pace since 2006 as it was driven by the Energy Mix Policy. There are several new plantations for biofuels crops (palm oil, sugarcane, cassava and *Jatropha curcas*) as well as developments in the processing industry associated with bioethanol and biodiesel. Up to 10^7 ha of land is to be allocated for biofuel crops alone by 2025. Of all biofuel crops, palm oil

has the highest productivity due to a high crop yield and a low production cost (Jupesta, 2011). Since palm oil is also important as a commodity for the food and cosmetics industries, the government plans to expand the current 9.7×10^6 ha of total palm oil plantations to 18×10^6 ha over the next decade, supposedly without disturbing forests (Jakarta Post, 2009). However, due to land constraints it appears that the expansion of palm oil production would lead to deforestation with consequent biodiversity loss, ecosystem changes and conflicts in society over water and land. Converting forest into palm oil plantations for biodiesel production would result in a carbon debt of ~610 t of CO_2 per ha which would take about 86 years to repay (Fargione et al., 2008). Currently, action is taken to stimulate cultivation of biofuels on degraded land, which means moving to a more sustainable production. The notion of degraded land, however, needs to be clearly defined to well distinguish between forest, peat land and degraded land (Jupesta et al., 2011).

As a third key impact, the REDD+ program, starting in 2009, conflicts with the land expansion for biofuels production area: it entails a moratorium on deforestation and promotes a reforestation program. The REDD+ program would drastically reduce emissions from peat fire and deforestation (LUCF). The Indonesian GHG abatement cost curve indicates that Indonesia has the potential to reduce GHG emissions by 2.3 $GtCO_2e$ by 2030, equal to 7% of global GHG emissions reduction required by the Intergovernmental Panel on Climate Change (Indonesia National Climate Change Commision, 2010). The funding of the REDD+ program has been decided so that Indonesia can benefit from 30 billion US$ of fast starting funds, as committed in December 2010 at the COP 16 of the UNFCCC in Cancun, Mexico (UNFCCC, 2010). The REDD+ national program needs to be strong and legally binding, as well as credible with quantitative objectives. In addition, it needs a transparent and accountable system for monitoring, reporting and verifying (MRV) the reductions in carbon emissions. Last but not least, it needs a payment system to compensate those who have incurred costs as a result of the REDD+ program. The accounting system is still under development and involves several stakeholders: the government of Indonesia, the government of Norway and other donor countries, technical experts from academia, indigenous people and funding agencies, such as the World Bank and ADB. Still, some argue that REDD+ payments may not be able to compete with the cost of oil palm production in forested areas, although they may provide sufficient incentives to shift development away from areas with high carbon content (World Bank, 2011). On peat land, which can store relatively much carbon, the breakeven carbon price is at 2–16 US$/t, lower than the carbon price of forest which reaches 10–33 US$/t. Under this scheme the REDD+ program is only able to avoid deforestation in the forested areas.

4. Overall evaluation

The Indonesian Energy Mix Policy is founded on two pillars: securing energy supply is important for maintaining and enhancing industrial growth, and increasing the share of renewable energy is a crucial element of a "green growth" path. While one may be critical of green growth in the case of developed (OECD) countries, emerging countries like Indonesia need growth for some time to meet basic development goals. Clear and significant progress of the REDD+ program could mean the development of an international framework for MRV and cooperation, such as the bilateral one between Norway and Indonesia. GHG emissions come to a large extent from peat fire and deforestation in Indonesia for the time being, but in the future it will increasingly originate from the electricity and transportation sectors. Thus, instead of enlarging the production of biofuels by land use change, which can cause deforestation, keeping those existing plantations and improving productivity through technology and innovation is more rational (Jupesta et al., 2011). Shares of renewable energy sources other than biofuels should increase based on their potential, such as for geothermal energy. In addition, a stringent policy regarding spatial planning is necessary to manage land allocation effectively and efficiently: to stop immediately peat fires and deforestation, and to allocate land to "flexible crops", i.e. which can be used as both food and fuel.

The government of Indonesia attempts to remove the barriers to green growth by eliminating costly energy subsidies in the near future, promoting cleaner energy sources, adopting payment for ecosystems services through the REDD+ program, and innovating its climate funding mechanisms. However some of the policies lead to conflicting outcomes. While the energy policy of 2006 seeks to

expand land for biofuel production, the climate policy of 2009 intends to implement a moratorium on deforestation. Regional autonomy and decentralization of economic and political power to provincial and local governments is crucial for this transition to pick up speed as these entities are better placed to address key social, economic and environmental concerns than the central government in Jakarta. The latter should limit its mandate to setting national standards and legal instruments for the transition to be in line with international climate economic and climate regimes. There is a need for coherence among existing policies to ensure a sustainability pathway for achieving green growth. Indonesia's case of sustainability transition could be a lesson for other developed and developing countries pursuing green growth. This issue is also relevant to another theme of the UNCSD, namely the institutional framework for sustainable development. This should involve systematic institutional governance by improving coordination among overseeing institutions.

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